Experimental Report on the School of Computer and Artificial Intelligence at Zhengzhou University

Grade：2024 Major：SOFTWARE ENGINEERING Class：GROUP-1

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Instructor：ALEX Score： 2026 year 10 month 10 date

Experimental Topic：Familiarization with Android Development Environment

1. Experimental Purpose

The main purposes of this experiment are:

- To gain familiarity with the components, configuration, and functionality of the Android development environment.
- To understand the structure of the Android SDK and its role in Android development.
- To learn how to create an Android project in Android Studio.
- To learn how to create and configure a virtual device (emulator) for testing Android applications.
- To learn how to build and run a pre-built Android sample and observe the results.

2. Experimental Content

In this experiment, the following tasks were carried out:

- Observing the Android SDK structure — exploring the folders and components of the Android SDK installed with Android Studio.
- Creating a new Android project using the Empty Activity template with Kotlin.
- Setting up a virtual device (Pixel 6 with API 37.2) using the Device Manager.
- Running a pre-built Android sample and verifying the output on the emulator.

3. Experimental Environment

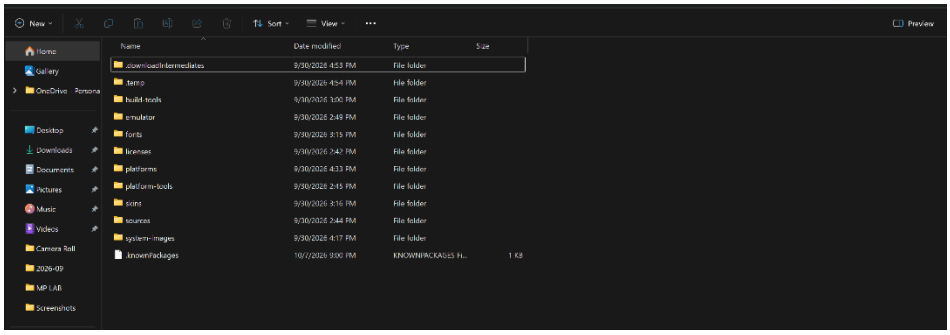
The experiment was performed on Windows 11 using Android Studio with Kotlin and Gradle. The project targets a minimum SDK of API 21 and was compiled with API 37.2. Testing was done on a Pixel 6 emulator running the Google APIs x86_64 system image. The Android SDK is installed at C:\Users\SAMSUNG\AppData\Local\Android\Sdk.

4. Procedure

The Android SDK is installed at: C:\Users\SAMSUNG\AppData\Local\Android\Sdk It contains the following key directories:

- build-tools/ — tools used by Gradle
- emulator/ — Android emulator resources
- platform-tools/ — tools like adb and fastboot
- platforms/ — installed Android platform versions
- sources/ — Android source code
- system-images/ — emulator system images
- licenses/, fonts/, skins/ — supporting resources

Figure 1: Android SDK Folder Structure

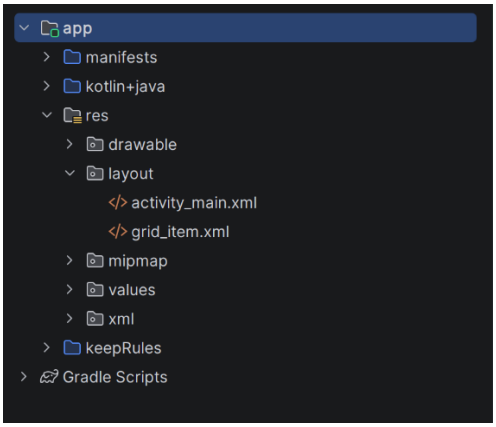


4.2 Creating a New Android Project A new Android project was created using the following steps:

1. File → New → New Project
2. Selected Empty Activity template
3. Project Name: Experiment1
4. Language: Kotlin
5. Minimum SDK: API 21 The generated project contains:

- MainActivity.kt — Kotlin source code
- AndroidManifest.xml — App configuration
- build.gradle.kts — Build configuration

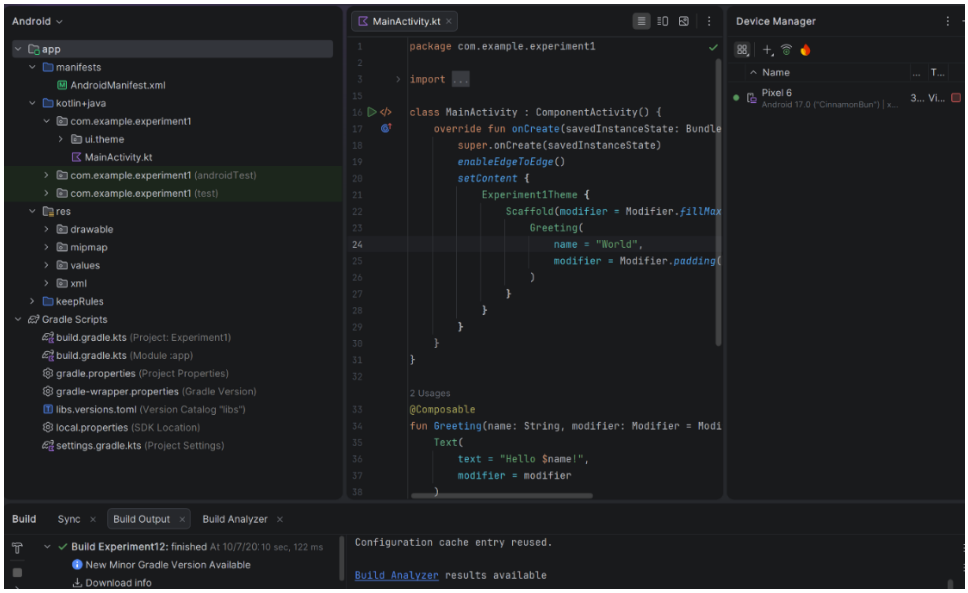
Figure 2: New Project Created



4.3 Setting Up a Virtual Device (Emulator) A virtual device (AVD) was created using the Device Manager:

1. Opened Device Manager → + → Create Virtual Device
2. Selected Pixel 6 (1080×2400, 420 dpi)
3. Chose system image: Google APIs Intel x86_64 (API 37.2)
4. Named the AVD → Finish 5. Launched the emulator successfully

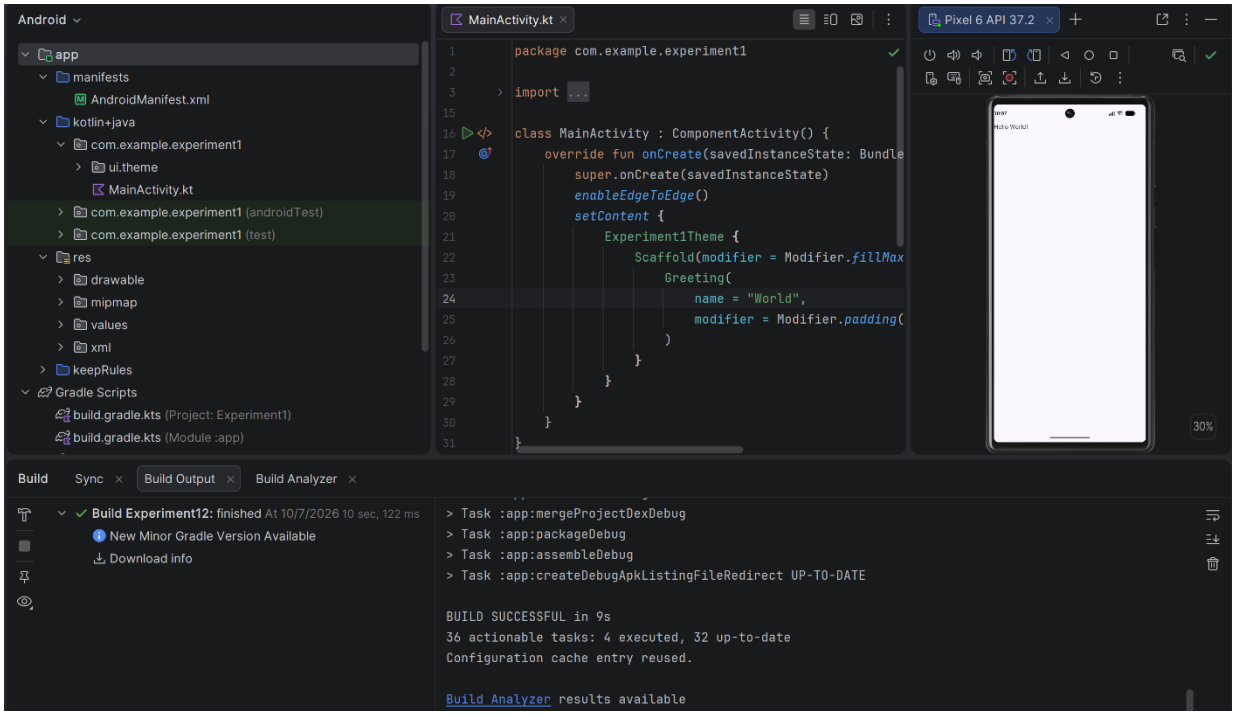
Figure 3: Virtual Device Setup



4.4 Running a Pre-built Android Sample

The pre-built Android application was built with Gradle and installed on the Pixel 6 emulator. The build completed successfully.

Figure 4: Build Successful



5. Results and Analysis

5.1 Application Output

After running the application on the Pixel 6 emulator, the sample app launched successfully, and all UI components were displayed correctly.

Figure 5: Application Output



The figure shows the "Hello World!" message displayed on the Pixel 6 emulator (API 37.2), confirming that the pre-built Android sample was successfully built, installed, and executed.

The Output confirmed the following:

- The Android SDK is properly configured and installed.
- A new project was created successfully with all required files.
- A virtual device (Pixel 6) was created and launched successfully.
- A pre-built Android sample was built and run without errors.

5.2 Analysis

From the results, the following observations were made:

- The Android SDK structure is well-organized, with separate folders for platforms, build tools, and system images.
- Android Studio provides an integrated environment for creating, building, and running Android projects.
- The Device Manager makes it easy to create and manage virtual devices.
- The emulator provides a safe environment for testing applications without needing a physical device.
- The Gradle build system efficiently compiled and installed the app on the emulator.

6. Summary

In this experiment, I successfully familiarized myself with the Android development environment. I learned how to observe the structure of the Android SDK, create a new Android project in Android Studio, set up and launch a virtual device using the Device Manager, and build and run a pre-built Android application on the emulator. Through this experiment, I gained practical knowledge about the Android development environment, the role of the Android SDK, the process of creating virtual devices, and running Android applications on an emulator. This knowledge provides a strong foundation for future experiments involving more advanced Android development.